

AI-Powered EHR Summarizer & Hospital Data Analytics Platform: A Generative-AI and Healthcare-Analytics Capstone

Capstone Project Report
Subject: Generative AI and Analytics for Clinics and Hospitalized Data

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1. Abstract

Hospitals and clinics generate two very different, and today largely disconnected, streams of digital information: the patient-level clinical narrative inside each Electronic Health Record (EHR), and the facility-level operational and clinical data that accumulates across every patient, department and encounter. Clinicians struggle to make sense of a single patient's fragmented longitudinal record in the limited time available per consultation, while hospital administrators and clinical managers lack a consolidated, data-driven view of trends across the whole patient population.

This project presents **Nexus Health**, a Hospital Intelligence Platform that combines a Generative AI **EHR Summarizer and Insights Generator** with a hospital-wide **analytics dashboard**. The GenAI layer uses a Large Language Model (LLM), clinical NLP and a **Retrieval-Augmented Generation (RAG)** pipeline to convert a patient's record into a concise, guideline-grounded clinical summary with risk and medication-conflict flags. The analytics layer reuses the same patient-encounter database to compute department-level and clinical-quality metrics for administrators.

The system uses a **LangGraph** workflow (Extract → Retrieve → Summarize → Flag) with a **FAISS** vector index for guideline retrieval, a **FastAPI** backend and a **React** frontend. The implemented prototype holds **50 patients**, **50 encounters** and **18 stored insight flags**, covers **6 departments**, and shows an average length of stay of **8.6 days**. Medication conflicts form the largest flag group, with **8 of the 18 flags (44.4%)**.

The final system integrates patient-level summarization, risk flagging, hospital-wide analytics and a live database viewer into a single platform. The generated summaries and flags are treated as **decision-support indicators requiring clinician review**, rather than a diagnosis or a final clinical decision.

2. Introduction

Electronic Health Records accumulate years of clinician notes, lab results, medication orders, discharge summaries and imaging reports for every patient. Before each consultation, a clinician must scroll through this fragmented, chronologically scattered record to reconstruct the patient's history, which is a slow and error-prone process. At the same time, hospital administrators need a bird's-eye view of the same underlying data: how many patients are admitted per department, which conditions are trending upward, and how lab-abnormality and medication-conflict rates are changing over time.

These are traditionally treated as two separate problems requiring two separate tools: a clinical NLP / Generative AI system for the clinician, and a Business Intelligence system for the administrator. This capstone treats them as two faces of the same data problem and proposes one integrated system, built jointly across the Generative AI and Analytics for Clinics and Hospitalized Data subjects, so that a single data pipeline and database serves both a clinician-facing summarization experience and an administrator-facing analytics dashboard.

The distinguishing concept of Nexus Health is therefore the combination of:

- Generative AI summarization of the longitudinal patient record
- RAG grounding with a FAISS index over clinical guidelines
- LangGraph agent workflow for ordered, traceable reasoning steps
- Automatic risk and medication-conflict flagging
- One shared database serving clinician and administrator views

- Hospital-wide analytics dashboard with clinical-quality metrics

2.1 Why Combine These Two Subjects Into One Capstone

- **Shared data source:** both subjects operate on the same patient, encounter and lab data. GenAI reasons over one patient's record at a time, while Analytics aggregates across all patients.
- **Shared database schema:** the schema designed for the GenAI module (patients, encounters, extracted_entities, insight_flags) is the natural fact table for the Analytics dashboards, so there is no duplicate data modelling effort.
- **Complementary outputs:** GenAI produces qualitative, explainable, per-patient insight; Analytics produces quantitative, population-level trend insight. Together they give a hospital both a micro and a macro view.
- **Realistic industry pattern:** real hospital IT systems pair a clinical-decision-support layer with a separate BI/analytics layer reading the same data warehouse. This project mirrors that pattern rather than inventing an artificial integration.

2.2 Literature Review Summary

A literature review was carried out as part of the GenAI subject experiments (Experiment 2), covering LLM-based clinical summarization, RAG-based clinical decision support, clinical NER pipelines, LangGraph-style agentic workflows and clinician trust in AI-generated summaries. The key works and the gaps identified are summarized below. The analytics component is motivated by the same underlying gap, a lack of a longitudinal, trend-aware view, but applied at the population level instead of the single-patient level.

Research Work	Method Used	Key Limitation Identified
Clinical Note Summarization Survey (2024)	Abstractive summarization (LLM)	No grounding in verified clinical guideline data
RAG for Clinical Decision Support (2024)	Retrieval-Augmented Generation	Text-only; no structured FHIR / hospital-scale analytics support
Clinical NER using scispaCy (2023)	Transformer-based entity extraction	No downstream verification or population-level aggregation
LangGraph Clinical Agents (2024)	Multi-step agent orchestration	Applied only to single-record summarization, not analytics
Clinician Trust in AI Summaries (2025)	Clinician-facing NLG review	No longitudinal, cross-patient trend comparison

Gap relevant to this capstone: none of the surveyed systems combine a grounded, patient-level GenAI summarization layer with a population-level analytics layer over the same data. This is the integration the capstone proposes.

3. Project Description

3.1 Problem Statement

Clinicians regularly review Electronic Health Records containing years of fragmented notes, lab results and medication histories before a visit, forcing them to scroll through disconnected data under significant time pressure. This increases the risk of overlooked trends and missed medication conflicts.

Existing chart-summary tools, where available, often generate ungrounded, single-encounter summaries without verifying values against clinical reference ranges or prior trends.

Simultaneously, hospital administrators and clinical managers have no consolidated, data-driven view of operational and clinical trends across the patient population. Admission patterns, department-wise load, disease prevalence and abnormal-lab or medication-conflict rates are either not tracked or tracked in isolated spreadsheets disconnected from the clinical system of record.

The proposed Nexus Health platform addresses this by generating grounded, patient-level summaries and risk flags for clinicians, while using the same stored data to power a hospital-wide analytics view for administrators, without replacing clinical or managerial judgement.

The system is designed around the following objectives:

GenAI subject objectives (patient-level)

1. Ingest structured FHIR data and unstructured or scanned clinical notes for a patient.
2. Integrate a large language model (GPT-4o / Med-Gemini) for clinical NLP and summarization.
3. Implement a RAG pipeline to ground every generated insight in verified clinical guideline data.
4. Automatically flag lab-value risk trends and medication conflicts.
5. Generate a concise, clinician-friendly pre-visit insight brief.
6. Enable longitudinal comparison of a patient's health values across encounters.

Analytics subject objectives (population / hospital-level)

1. Design a hospital-wide analytical data model on top of the same patient and encounter schema.
2. Build descriptive dashboards for admissions, department load and length-of-stay trends.
3. Compute aggregate clinical-quality metrics, such as abnormal-lab rate and medication-conflict rate, reusing the insight_flags table.
4. Build a simple predictive model (logistic regression or decision tree) for patient-readmission risk or short-term bed-demand forecasting.
5. Present findings through an interactive dashboard for hospital administrators and clinical managers.

Combined objective: to demonstrate, through one working platform, how a Generative-AI clinical layer and a data-analytics operational layer can share a single data foundation to serve two different stakeholders, the clinician and the administrator, responsibly and without replacing human judgement.

3.2 System Architecture

The Nexus Health platform is organized into a shared data layer, two processing branches, an API layer and a presentation layer.

The **data layer** receives hospital and EHR data (records, encounters, notes, labs and medications) through an ETL step (Extract, Transform, Load) and stores it in a database holding patients, encounters and insight flags. From this database the system splits into two branches. **Branch A (patient-level AI)** uses LangGraph, RAG and an LLM to produce a structured JSON output containing the summary, flags and insights. **Branch B (hospital analytics)** uses Pandas aggregation to produce a JSON response containing counts, average stay and department statistics.

Both branches feed a **FastAPI backend** that exposes REST endpoints returning JSON. The **React frontend** calls these endpoints with Axios, keeps the results in React state and renders them in UI components: a **clinical dashboard** (summary, risk flags, insights) for the clinician, and an **analytics dashboard** (Plotly charts, department statistics) for the administrator. Branch A also saves the flags it detects back to the database, so the analytics branch can count them.

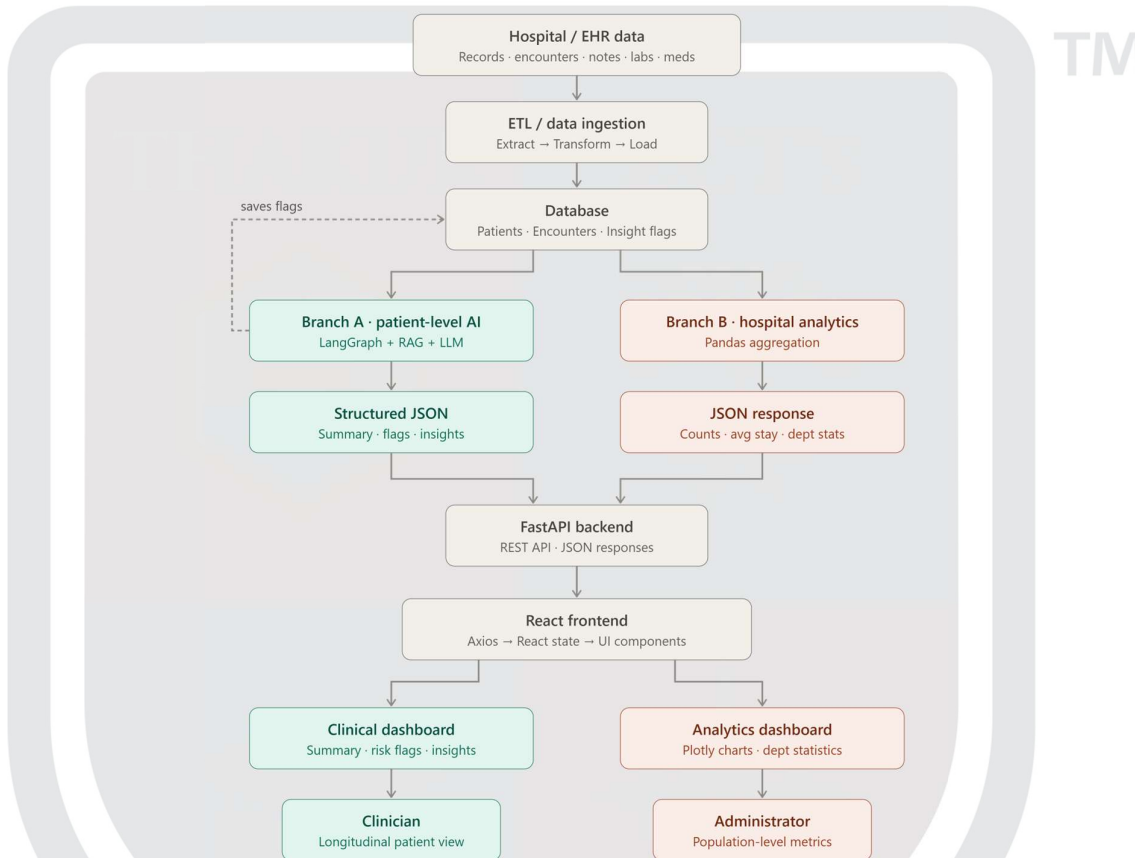


Fig. 1 System Overview of the AI-Powered EHR Summarizer & Hospital Analytics Platform

The data and control flow of the platform is as follows:

1. A patient's structured FHIR data and/or scanned notes are ingested (GenAI).
2. Clinical NLP extracts structured entities; the LangGraph pipeline retrieves grounded guideline data through FAISS RAG and generates a summary and risk flags (GenAI).
3. All extracted entities, flags and summaries are persisted to the shared database.
4. The Analytics ETL layer aggregates records across all patients into fact tables and views (Analytics).
5. KPI dashboards compute admission counts, length-of-stay, department distribution and flag-rate metrics from the aggregated data (Analytics).
6. The predictive model, once built, will retrain periodically and score current inpatients for readmission risk (Analytics, planned).

7. Clinicians view the clinical dashboard and administrators view the analytics dashboard, both backed by the same source of truth.

3.3 Methodology and Design

Each component of the system was designed around a specific function and validated before integration.

The first stage is **ingestion and entity extraction**, where a patient's structured data and clinical notes are read and the clinically relevant entities (conditions, medications, vital signs such as blood pressure, lab values and their units) are extracted. The second stage is **retrieval**, where the extracted patient context is used to search a clinical guideline knowledge base. The guidelines are converted into vector embeddings by an embedding model and stored in a **FAISS** index; a similarity search returns the most relevant guideline passages. The knowledge base is used for retrieval only.

The third stage is **summarization**. The patient context and the retrieved guideline passages are passed together to the LLM, which generates a grounded longitudinal clinical summary. The fourth stage is **flagging**, where the structured output (summary, risk flags, insights) is checked for risk conditions, such as blood pressure above the guideline threshold or a conflicting medication pair, and the resulting flags are persisted to the database. The **LangGraph** workflow controls the order of these four steps (Extract → Retrieve → Summarize → Flag), and the execution path is shown to the clinician as a trace.

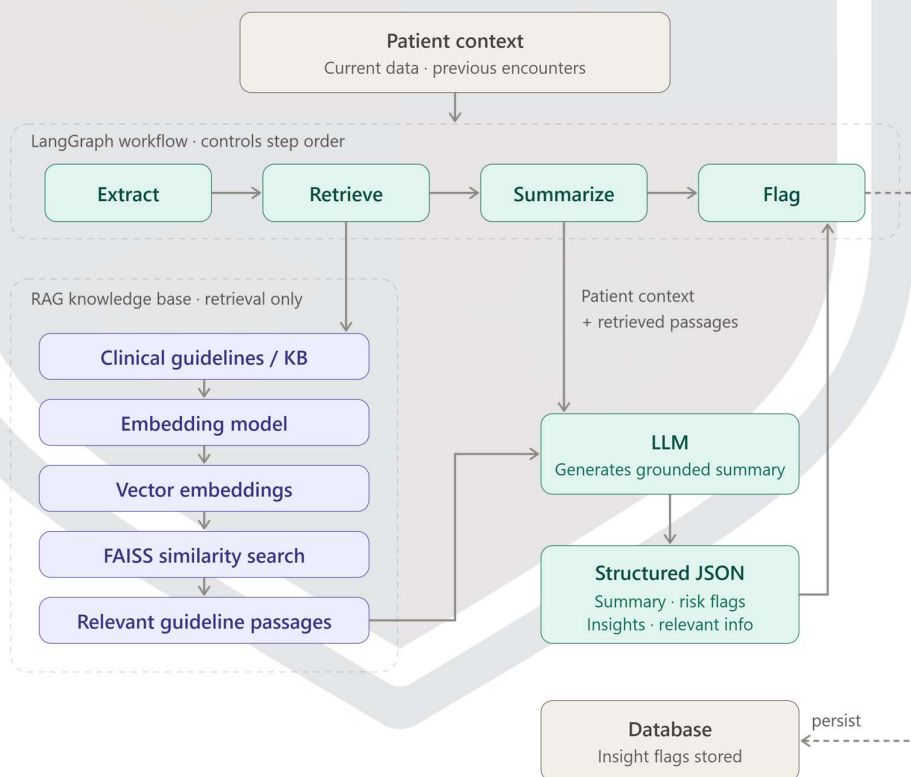


Fig. 2 Patient-Level AI Branch: LangGraph Workflow with RAG Knowledge Base

The final stages are **analytics** and **presentation**. The analytics branch aggregates the shared tables with Pandas to produce population-level counts, average length of stay, department distribution and flag-

category distribution, which are shown on the analytics dashboard. The two subsystems exchange data as shown below.

Data Produced By	Consumed By	Purpose
GenAI: extracted_entities, insight_flags (per patient)	Analytics: aggregate KPI dashboard	Turns per-patient risk flags into hospital-wide quality metrics
GenAI: encounters table (per patient)	Analytics: admissions and length-of-stay charts	Turns individual encounters into population-level operational trends
Analytics: readmission-risk model score (planned)	GenAI: pre-visit insight brief (optional extension)	Surfaces a population-trained risk score back into the clinician-facing brief

3.4 Implementation

The patient-level AI branch is implemented as a LangGraph agent pipeline orchestrated by a FastAPI backend. For a given patient ID, the **Clinician AI Portal** runs the pipeline and displays three things side by side: a **Longitudinal Clinical Summary**, a panel of **Detected Risk Flags & Conflicts** with the grounded guideline source for each flag, and a **LangGraph Trace** showing the agent execution path and the guideline passages retrieved by FAISS. Showing the retrieved guidelines next to the output lets the clinician check where each flag came from.

The guideline knowledge base used in the prototype includes rules such as: systolic blood pressure over 140 or diastolic over 90 is considered high blood pressure (Guideline A); taking Aspirin and Ibuprofen together can increase the risk of gastrointestinal bleeding (Guideline B); and the maximum daily dose of Paracetamol is 4000 mg to prevent hepatotoxicity (Guideline G).

The analytics component aggregates the shared database using Pandas and presents the results on the **Hospital Analytics** page using Plotly charts: summary tiles, a patient-distribution bar chart by department and a donut chart of insight-flag categories. A **Live SQL Viewer** page shows the connected database, the row count of each table and the result of a sample query, so that the data behind both views can be inspected directly.

For development, the prototype is connected to a local SQLite database (hospital.db) holding the patients, encounters and insight_flags tables. The target design uses a managed PostgreSQL instance with the same schema.

3.5 Technology Stack

Layer	Technology
GenAI Backend & Orchestration	Python 3.11+, FastAPI, LangGraph, GPT-4o / Med-Gemini API
Clinical NLP	scispaCy, transformer-based NER, HL7 FHIR library
RAG / Retrieval	FAISS vector index, sentence-embedding models
Frontend	React / Next.js, Axios
Analytics & Data Science	Python (Pandas, scikit-learn), SQL
Analytics Dashboard	Plotly (Power BI / Plotly Dash considered)
Database	SQLite (prototype), PostgreSQL (target design)

Layer	Technology
DevOps / Deployment	Docker, GitHub Actions (CI/CD), Vercel (frontend), Render / AWS (backend)
Tools	Git & GitHub, Postman, Draw.io / Lucidchart, Figma

4. Results and Discussions

4.1 Clinician AI Portal Results

The Clinician AI Portal generates a longitudinal summary, a set of risk flags and an execution trace for a patient ID. Two cases from the prototype are shown below.

Patient ID	Condition	Key Inputs	Output
1	Cancer	BP 119/89; Paracetamol	No risks or medication conflicts detected
31	Asthma	High blood pressure; Aspirin and Ibuprofen	2 flags: Medication Conflict (Guideline B), High Blood Pressure (Guideline A)

For patient 1, the blood pressure of 119/89 is below the Guideline A threshold (above 140 systolic or above 90 diastolic) and Paracetamol is within the retrieved dose guideline, so no flag is raised (Fig. 3). For patient 31, the portal raises a **Medication Conflict** flag grounded in Guideline B and a **High Blood Pressure** flag grounded in Guideline A (Fig. 4).

Fig. 3 Clinician AI Portal: Patient 1, No Risk Flags Detected

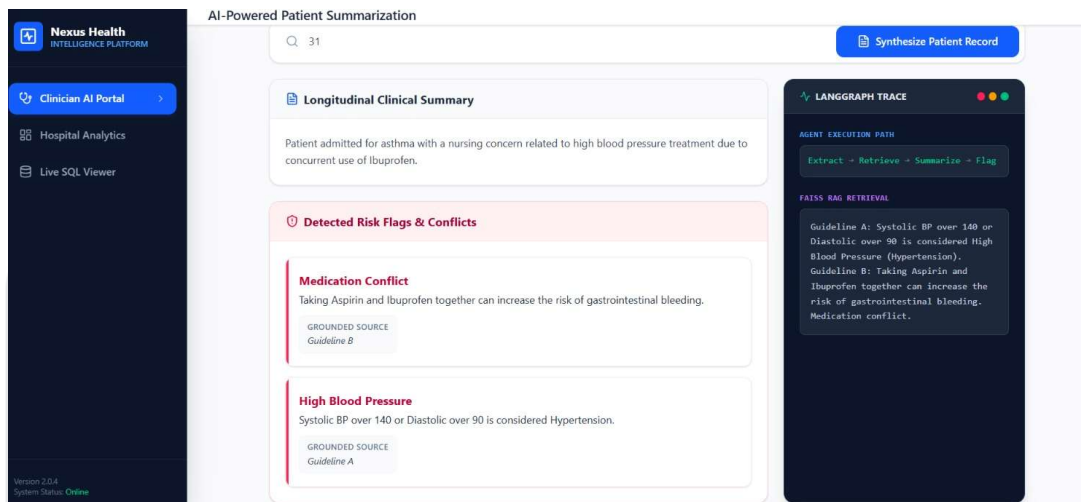


Fig. 4 Clinician AI Portal: Patient 31, Medication Conflict and High Blood Pressure Flags

Each flag carries a grounded source, and the retrieved guideline text is visible in the trace panel. The summary is a short natural-language text and is meant to be read together with the flags and sources; for example, the summary for patient 31 mentions Ibuprofen only, while Guideline B concerns Aspirin and Ibuprofen together. This is why the output is treated as decision support that a clinician reviews, not as a final record.

4.2 Hospital Analytics Results

The Hospital Analytics page summarizes the stored data for the whole patient population.

Metric	Result
Total admitted patients	50
Average length of stay	8.6 days
Clinical risk flags	18
Active departments	6



Fig. 5 Hospital Population Analytics Dashboard

Patient distribution by department. Asthma has the largest share of admitted patients, followed by Hypertension.

Department	Patients	Share of 50 patients
Asthma	12	24%
Hypertension	10	20%
Cancer	9	18%
Diabetes	8	16%
Obesity	7	14%
Arthritis	4	8%
Total	50	100%

Clinical quality: insight-flag categories. The donut chart splits the 18 stored flags by category.

Flag Category	Flags	Share of 18 flags
Medication Conflict	8	44.4%
Hypertension	3	16.7%
High Blood Pressure, Monitoring Required, Potential Risk, Risk, Risk Factor, Risk Type, Type (1 each)	7	5.56% each
Total	18	100%

Medication conflicts are the largest flag group. The remaining categories show that the flag labels are not yet standardized: "Hypertension" and "High Blood Pressure" describe the same condition, and generic labels such as "Risk", "Risk Type" and "Type" carry little meaning. If these are merged, the blood-pressure related flags would be at least 4 of 18 (22.2%). A fixed set of flag types should therefore be applied before the flag-rate KPIs are used for quality monitoring.

5. Conclusion

This project demonstrates the feasibility of combining a **guideline-grounded Generative AI summarizer** with a **hospital-wide analytics dashboard** on one shared database.

The Nexus Health prototype generates a longitudinal clinical summary, grounded risk flags and a visible agent trace for a patient, and it turns the stored flags and encounters of 50 patients into department and clinical-quality metrics for administrators. The implemented dashboards show 50 patients, 18 insight flags, 6 departments and an average length of stay of 8.6 days.

The major contribution of the project is the integration of two complementary capabilities:

The GenAI layer answers "What does this patient's record show, and what needs attention?" while the Analytics layer answers "What is happening across the hospital?"

The system should be considered a **prototype decision-support platform**. A generated summary or flag does not replace clinical judgement, and a population-level metric does not replace managerial review. The readmission-risk model and several validation tests are still planned.

5.1 Advantages

- One shared database removes duplicate data-modelling work across the two subjects.
- Clinician-facing (micro) and administrator-facing (macro) views stay consistent, since both read the same source of truth.
- RAG grounding reduces the risk of hallucination on the GenAI side.
- Evidence-based hospital planning is enabled by the Analytics side.
- The modular architecture allows each subsystem to be developed, tested and evaluated independently while still integrating cleanly.

5.2 Applications

- Clinician pre-visit preparation and chart review (GenAI).
- Hospital administration and capacity planning (Analytics).
- Clinical quality monitoring, tracking abnormal-lab and medication-conflict rates over time (Analytics).
- Department-level resource allocation based on admission and length-of-stay trends (Analytics).
- Early-warning readmission-risk screening to support discharge planning (Analytics, planned).

5.3 Limitations

- Performance of both subsystems depends on the quality and volume of available data. The prototype uses 50 patients and 50 encounters.
- The predictive analytics model will provide a risk score, not a diagnosis, and requires clinical validation before any real deployment.
- AI-generated summaries and analytics outputs are estimates and must be reviewed by qualified clinical and administrative staff.
- Flag categories are not yet standardized, which limits the accuracy of flag-rate KPIs.

6. Future Scope

The current system can be extended in several directions:

1. **Readmission-risk model** - Complete the planned classifier (logistic regression or decision tree) and evaluate it on a held-out sample.
2. **Closed-loop risk score** - Feed the readmission-risk score back into the GenAI pre-visit brief.
3. **Standardized flag taxonomy** - Map all flag labels to a fixed set of types so that flag-rate KPIs are reliable.
4. **Explicit verification step** - Add a verification stage that checks extracted values against clinical reference ranges before retrieval and summarization.
5. **Forecasting** - Extend predictive analytics to bed-demand and staffing forecasts using time-series models.
6. **More record types** - Support imaging and pathology records in both subsystems.

7. Annexure

7.1 Shared Database Schema

The schema was originally designed for the GenAI subsystem and is extended with two lightweight tables (admissions and readmission_scores) to support the Analytics subsystem, avoiding duplication of patient and encounter data.

Table	Key Fields	Used By
patients	patient_id (PK), name, dob, primary_specialty_context	GenAI + Analytics
encounters	encounter_id (PK), patient_id (FK), encounter_type, encounter_date, provider_id	GenAI + Analytics
extracted_entities	entity_id (PK), encounter_id (FK), entity_type, value, unit, reference_range	GenAI + Analytics
insight_flags	flag_id (PK), entity_id (FK), flag_type, status, explanation	GenAI + Analytics
summaries	summary_id (PK), patient_id (FK), summary_text, pre_visit_brief	GenAI
query_history	message_id (PK), patient_id (FK), clinician_id (FK), role, message, timestamp	GenAI
reference_knowledge_meta	doc_id (PK), source, topic, faiss_index_ref	GenAI
admissions (new)	admission_id (PK), encounter_id (FK), department, ward/bed_id, admit_date, discharge_date, billing_category	Analytics
readmission_scores (new)	score_id (PK), patient_id (FK), computed_date, risk_score, model_version	Analytics

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